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VRMotion — An Invention to Innovation Case-study



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CONTINUOUS INNOVATION

It was almost 8:45 am as we pulled into the Birrfeld airstrip just outside Zurich, Switzerland. I was meeting Fabian Riesen (Fabi), who had casually offered to take me flying over the Swiss Alps the week before. As this was my first visit to Switzerland, I jumped on the offer.

As we walked out towards the runway, I saw Fabi with his flying instructor, Heinz, by the plane. Fabi is a Cisco engineer and an avid supporter of “Running Lean,” which is how we first met over a Skype call. We had spoken

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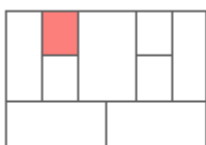


a few times before, but this was the first time we met in person.

After the flight check, we were on our way and up in the air. The weather couldn't have been better. Five minutes later, we were taking in breathtaking views of the Alps. Not a bad way to be introduced to the Alps...

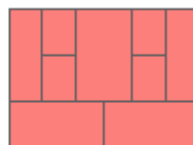
I drove back with Fabi. That's when he told me that he also had a flight simulator in his living room—one he had built himself as a fun side project. It had caused quite a bit of excitement, first with the neighbors and then with the local media. How does one build a flight simulator in their spare time?

Over the next year and a half, I have watched Fabi turn his passion side project from an invention into an innovation—one that is starting to get real traction.



Invention

versus



Innovation

The Idea Spark

Fabi's passion for flying goes back to his childhood when he and his friend Ralph dreamed of being fighter pilots. Fabi's dream was shattered early after he got glasses in

kindergarten, but Ralph did go on to become a fighter pilot.



This passion would be rekindled several years later in university. Fabi and his friends, Roger and Schneidy, decided to attempt building a flight simulator as part of their diploma work. The tech for 3D simulation was not quite there yet in 1998, so they built a train simulator instead. Their project was novel enough to attract 35,000 visitors, garner media attention, and recruit them by Cisco.

Around that time, Fabi also found out that he wasn't unfit to fly (despite his glasses), so he enrolled in flying school. There he discovered that runways in Switzerland are much shorter than average and require a "fly by your seats" approach. This re-triggered his interest in flight simulators, and he started looking around for what was available...

The prototype (2014)

There was a 6DOF (six degrees of freedom) motion platform from a German vendor for \$85K. A cheaper 4DOF simulator ran for \$35K. Even if he managed to buy one, these things were massive, required disassembly (voiding their warranty), and 400V of power.

Building a 6DOF simulator required six linear actuator motors. Each ran \$4,000 a piece. While cheaper, it was still way over his budget. He turned to scour the enthusiast community instead for other options.

This led to several dead-ends until he accidentally stumbled into Richard Elliot's youtube channel (the famous Saxophone player), where he found a video on an early prototype of a flight simulator that Richard was building. Following this trail led him to the engineer behind the project—Thanos Kontogiannis. Thanos had built an electronic frequency controller that worked with regular AC motors. Compared to linear actuators, an AC motor runs for ~\$100. Even with some extra components, this was still an order of magnitude cheaper than the alternatives.

There were still technical risks and overheating concerns as no one had successfully used these motors to control a flight simulator yet. But this was still a promising direction, so Fabi got his hands on just one of these controllers and started testing it with an AC motor and a crank arm. It passed the test.

His next task was building a frame for the simulator. Since he didn't have formal training in mechanical engineering, instead of using a 3D software tool, he opted for physical prototyping—with wood. Not only was this a cheaper option, but it allowed him to iterate more quickly on the model until he got all the motors correctly positioned and the frame stable. In a fail scenario, the wood would snap, which was the feedback loop that prompted him to reposition the motors and/or reprogram the simulator to avoid this condition.

Once he was satisfied with his model, he open-sourced all his specifications on his blog (<http://6dofabi.blogspot.ch/>) as a way to give back to the community.



From concept to prototype

Early traction

As he was invading the family living room with this project, he got everyone involved. His kids helped him with wiring, learned some coding, and got behind the project. Once the simulator was functional, word quickly got out in his neighborhood.



Early tests

Other kids (and their parents) started coming by to test-drive the simulator. Word of mouth spread quickly, and Fabi was contacted by Swiss television did a report on the simulator, getting him free publicity.

Version 2.0 and Team Formation (2015)

Shortly after that, Fabi started an eMBA course and convinced his Professor to use the flight simulator for the class project. There were two members in the class, Adi S (a mechanical engineer) and Adi F (an electronics engineer), who volunteered to help create and fund the building of a 2.0 version. This was a low-risk investment as the simulator was already functional. In the best case, they'd have an improved design.

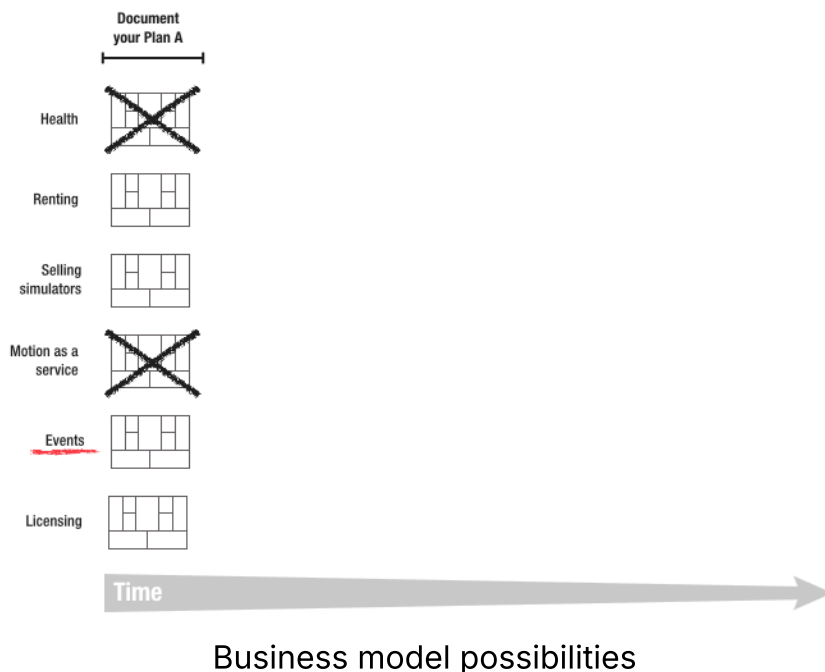
Otherwise, they'd end up with an identical clone of the current version. This is also when Fabi ran into Sebastian (software engineer), who would make the 4th member of the core team.



Standing from left to right; Eyüb U., Dominik L., Adi S., Roger, Carmen N., Fabi. Sitting Adi F | Version 2.0

Business Model Brainstorming

During the eMBA course, the class started applying Running Lean and brainstormed several possible business models.

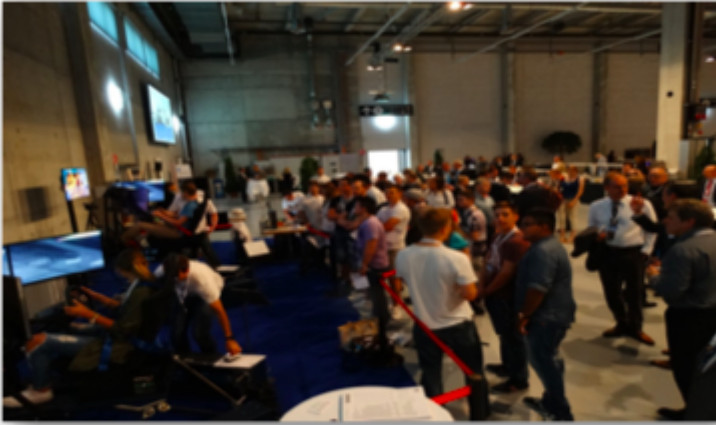


The goal of brainstorming is casting a wide net to consider all kinds of possibilities—even some that might be outright crazy.

The most promising one at the time was running events which are what they decided to pursue first.

The Events business model

They quickly secured two customers:
SwissMem and Cisco.



SwissMem event

The value proposition for SwissMem was creating an attention-driving booth at their public events for lead generation. Cisco used the simulator during offsite all-hands employee events for team building.

The problem with a fight simulator app is that it typically takes up to 10 mins per person. At events with long lines, you need something faster. So the team started exploring different apps and settled on adding a car racing app. This led them down the path of integrating with game engines (unity and unreal engine) that opened up a whole array of other applications.



Game engine integration

The team verified their assumptions by measuring queue lines and found 2 minutes per ride to be the sweet spot for maximizing entertainment with the sponsor's business model goal.

They also started tinkering with Go-Pro cameras and virtual reality glasses shortly after. And eventually built a prototype that could remote control a real drone using real-time motion.



Real-time Motion

Building a more robust platform through partnering and discovering their unfair advantage

While the team had made a lot of progress, they had “hacked their way” thus far,

especially on the software side (using Visual Basic), and decided to invest in building a more robust platform.

They approached ICOM institute from HSR university and made a deal where the university would help them rewrite the software and get them CE certified in exchange for two simulators that they could buy at cost and use towards further research and development.

This partnership (led by Dr prof. Guido Schuster) not only resulted in a more robust and sellable platform but also a few patents on their motion algorithms. This is where they built the capability to extract motion out of any video that could then be played on the simulator.

Current Status

Recall that the project had started around a vision of building a training simulator for pilots. The team realized that part of the story when they sold their first simulator to the FGZO flight school.

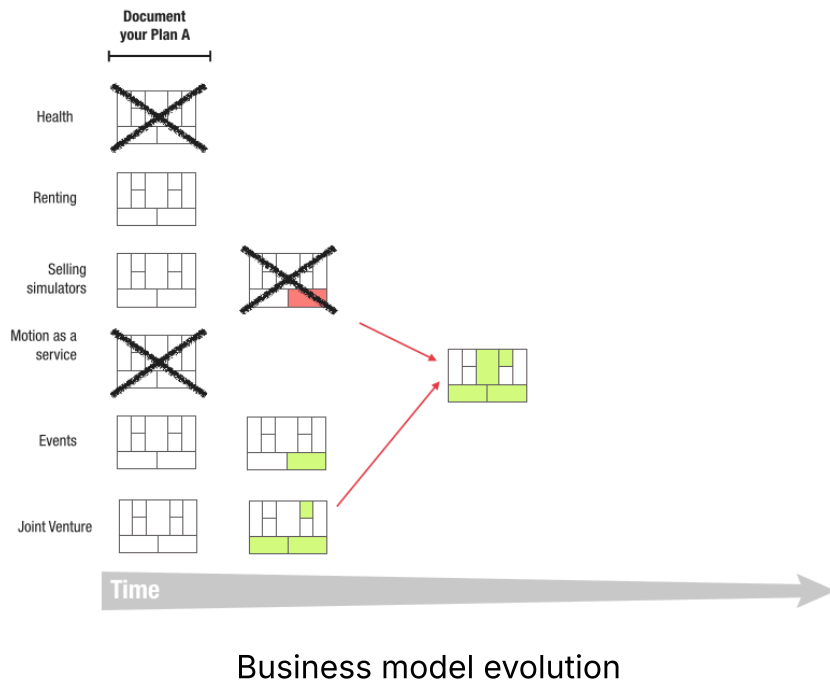


Cessna-172 Simulator at airfield Speck FGZO

However, they learned that building and selling simulators from Switzerland, where labor costs are astronomical, wasn't going to scale—especially since they had open-sourced all their original plans along the way.

Their real unfair advantage didn't lie in the open-sourced community version of the simulator where they started. Still, in the end-to-end solution, they iterated through a set of strategic partnerships and joint ventures along the way.

Through these partnerships, VRMotion made several new “inventions” (that they control and own) that are key to making their simulators work with VR glasses—an area that, not surprisingly, the team is heavily betting on.



What I love about this case study is that it showcases the journey from idea through technical feasibility (can we build this?) to business model innovation (will anyone care?). It also highlights how unfair advantages can be discovered and built over time, which I know is a box on the Lean Canvas that keeps many entrepreneurs up at night.

The innovation journey is seldom a straight shot but one where a systematic approach to tackling the right risks at the right time increases the odds of coming out at the other end.

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